CLASS 8 · CBSE · GANITA PRAKASH PART 2 · FORMULA BANK

Chapter 4: Exploring Some Geometric Themes

Every formula and result of the chapter on one sheet

1	Sierpinski Carpet, squares	$R_n = 8^n$
2	Sierpinski Carpet, holes	$H_n = 1 + 8 + \ldots + 8^{n-1}$
3	Sierpinski Carpet, area	$\left(\frac{8}{9}\right)^n$
4	Sierpinski Triangle, triangles	3^n
5	Sierpinski Triangle, area	$\left(\frac{3}{4}\right)^n$
6	Koch Snowflake, sides	3×4^n
7	Koch Snowflake, perimeter	$3 \times \left(\frac{4}{3}\right)^n$
8	Prism on an n-gon	$n + 2$ faces, $3n$ edges, $2n$ vertices
9	Pyramid on an n-gon	$n + 1$ faces, $2n$ edges, $n + 1$ vertices
10	Euler's relation	$F + V - E = 2$
11	Nets of a cube	11
12	Net of a cylinder	Rectangle h by $2\pi r$ and two circles
13	Shortest path, corner to opposite corner	Smallest of $\sqrt{(l+b)^2 + h^2}$, $\sqrt{(l+h)^2 + b^2}$, $\sqrt{(b+h)^2 + l^2}$
14	Projection of a segment	Length $\leq$ actual length
15	Three views	Front (vertical plane), top (horizontal plane), side (side plane)